Z-Test

1. Why Do We Need Z-Test?

In real life, we often want to check whether a claim about data is true or not.

But usually we cannot check the entire population, so we collect a sample.

The problem is:

Sample values can be different from the population just because of randomness.

So the question becomes:

Is the difference real or just random variation?

Z-test helps answer this question.

Real Life Example 1 (Very Easy)

A coaching institute claims:

Average marks of students = 70

A sample of 40 students is tested.

Average marks in sample = 74

Now the question is:

Did the institute really improve performance?

OR

Did the sample just randomly get slightly higher marks?

This is exactly what Z-test checks.

Real Life Example 2 (Industry Example)

A milk company claims:

Average milk packet weight = 1 liter

Quality team tests 50 packets.

Average weight = 1.03 liter

Question:

Is the machine actually producing more milk?

OR

Is this small difference due to random variation?

Z-test helps decide.

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2. Concept Behind Z-Test

Every dataset has natural variation.

Even if population mean is 70, sample means could be:

68
69
71
72
74

This variation happens due to random sampling.

So we measure:

How far the sample mean is from the population mean.

Why Standard Deviation Matters

Standard deviation shows natural spread of data.

If spread is large → difference may be normal.

If spread is small → difference may be important.

Z-test measures difference relative to the spread of data.

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3. Z-Test Formula

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Where

x̄ = sample mean
μ = population mean
σ = population standard deviation
n = sample size

Meaning of Each Part

(x̄ − μ)

Difference between sample mean and population mean.

This shows how much the sample deviates.

σ / √n

This represents standard error.

It measures expected variation of sample means.

Entire Formula

Z-score measures:

How large the difference is compared to expected random variation.

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4. Mathematical Example

Population mean marks

μ = 70

Population standard deviation

σ = 10

Sample size

n = 40

Sample mean

x̄ = 74

Step 1

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Step 2

Substitute values

$$Z = \frac{74 - 70}{10/\sqrt{40}}$$

Step 3

$$\sqrt{40} = 6.32$$
$$10/6.32 = 1.58$$

Step 4

$$Z = 4/1.58$$
$$Z = 2.53$$

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5. Interpretation

Normal distribution rule:

95% data lies between

−1.96 to +1.96

If Z-score lies outside this range, difference is statistically significant.

Here

Z = 2.53

Since

2.53 > 1.96

Conclusion:

The difference is too large to be random.

Null hypothesis is rejected.

Meaning in Simple Words

The average marks of the sample are significantly higher than expected.

So the original assumption that average marks = 70 may be incorrect.

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6. Graphical Meaning

Normal distribution curve

```
          Mean
     |
  ______|______
 /        \
/          \
-1.96       +1.96
```

Middle area = normal variation.

Outside area = unlikely event.

Z-test checks whether sample result lies in unlikely region.

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7. Why Z-Test is Important in Data Science

Z-tests are used in:

A/B Testing

Example:

Testing which website design gets more clicks.

Product Experiments

Testing whether a new feature improves engagement.

Quality Control

Checking whether products meet standard specifications.

Fraud Detection

Detecting unusual transactions.

Idea of Z Score

Z-score tells:

How many standard deviations away a value is from the mean.

Example

Mean marks = 70
Standard deviation = 10

If a value is:

80

$$Z = \frac{80 - 70}{10} = 1$$

Meaning:

80 is 1 standard deviation above the mean.

1. Introduction to Hypothesis Testing

Hypothesis testing is a statistical method used to determine whether there is sufficient evidence in a sample dataset to support a claim about a population parameter.

It allows making statistical decisions using sample data.

Example

A company claims that the **average battery life of a smartphone is 10 hours**.

To verify this claim, a sample of phones is tested.
Hypothesis testing helps determine whether the observed sample result supports or contradicts the claim.

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2. Statistical Hypothesis

A statistical hypothesis is an assumption about a population parameter.

Two hypotheses are considered.

Null Hypothesis (H_0)

The null hypothesis represents the **default assumption** that there is no difference or no effect.

Example

H_0: $\mu = 50$

Meaning the population mean is 50.

Alternative Hypothesis (H_1)

The alternative hypothesis represents the **opposite claim** of the null hypothesis.

Example

H_1: $\mu \neq 50$

Meaning the population mean is not equal to 50.

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3. What is a Z-Test

A **Z-Test** is a statistical test used to determine whether the difference between a sample mean and population mean is statistically significant.

It is applied when:

- Sample size is large ($n \geq 30$)
- Population standard deviation is known
- Data approximately follows normal distribution

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4. Mathematical Formula of Z-Test

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Where

$\bar{x}$ = Sample mean
μ = Population mean
σ = Population standard deviation
n = Sample size

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5. Meaning of the Formula

The Z-score represents how many **standard deviations the sample mean deviates from the population mean**.

If this deviation is large, the difference is statistically significant.

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6. Step-by-Step Mathematical Implementation

Given

Population Mean

$\mu = 70$

Population Standard Deviation

$\sigma = 10$

Sample Size

$n = 40$

Sample Mean

$\bar{x} = 74$

Step 1: Define Hypotheses

Null Hypothesis

$H_o : \mu = 70$

Alternative Hypothesis

$H_1 : \mu \neq 70$

Step 2: Apply Z-Test Formula

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Substitute values

$$Z = \frac{74 - 70}{10/\sqrt{40}}$$

Step 3: Solve Denominator

$$\sqrt{40} = 6.32$$
$$10/6.32 = 1.58$$

Step 4: Calculate Z Score

$$Z = 4/1.58$$
$$Z = 2.53$$

Step 5: Compare with Critical Value

For significance level

$\alpha = 0.05$

Critical Z-value = ±1.96

Since

$|2.53| > 1.96$

Step 6: Decision

Reject Null Hypothesis.

Conclusion:

Sample evidence suggests that the population mean is significantly different from 70.

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7. Types of Z-Test

1 One-Tailed Test

Used when the alternative hypothesis specifies a direction.

Example

H_1: $\mu > 70$

or

H_1: $\mu < 70$

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2 Two-Tailed Test

Used when the alternative hypothesis checks difference in both directions.

Example

H_1: $\mu \neq 70$

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8. Z-Score

Z-score measures the distance of a data point from the mean in terms of standard deviations.

Formula

$$Z = \frac{x - \mu}{\sigma}$$

Example

Mean height

$\mu = 170$

Standard deviation

$\sigma = 5$

Observed height

$x = 180$

$$Z = \frac{180 - 170}{5}$$
$$Z = 2$$

Meaning the value is **2 standard deviations above the mean**.

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9. Graphical Interpretation

Z-test follows the **normal distribution curve**.

```
                Mean
        |
    ______|_______
   /          \
  /            \
 -1.96         +1.96
```

Region between -1.96 and +1.96 → Accept H_0
Region outside → Reject H_0

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10. Applications of Z-Test

Z-test is widely used in:

Quality Control

Testing whether manufactured products meet standard specifications.

Medical Research

Testing effectiveness of medicines.

A/B Testing

Comparing two versions of websites.

Data Science

Evaluating statistical significance in experiments.